

Kushagra Tiwari

+91-8127328573 • kushagrace.techai@gmail.com • /kushagra-tiwari-a695a0268/ • /Kushagrace2803

Summary

AI/ML Engineer specializing in end-to-end PyTorch/TensorFlow pipelines and scalable REST APIs. Proven track record in training custom GPT architectures, architecting intelligent multi-LLM routing systems, and optimizing local SLMs. Expert in designing advanced RAG pipelines and deploying edge-AI/multimodal solutions

Education

•Pranveer Singh Institute of Technology, Kanpur

Bachelor of Technology in Computer Science and Engineering (AI)

2022 – 2026

CGPA: 7.7 / 10.0

Professional Experience

•XprtCode

Artificial Intelligence Intern

July 2025 – Nov 2025

Remote

- Built RAG chatbot using **LangChain** and OpenAI, reducing query resolution time by **40%**.
- Designed and deployed RESTful APIs using **Flask** to serve ML models, ensuring scalable backend integration.
- Optimized data preprocessing pipelines using **Pandas** and **NumPy** for datasets exceeding 100,000 records, improving training efficiency.

Projects

•Elysium AI

GenAI Platform with Custom GPT-style LLM and Multi-Model Routing

GitHub

Oct 2025 – Present

- Architected **Elysium LLM** from scratch implementing custom tokenization, embeddings, causal self-attention, causal masking, and stacked transformer blocks in PyTorch.
- Engineered an intelligent **multi-LLM router** to dynamically switch execution between Groq, Gemini, Ollama.
- Deployed MultiModel GenAI tools (Quiz Generator, Summarizer, Text-to-Video) to automate scalable content creation.
- **Technologies:** PyTorch, Flask, LangChain, Hugging Face Transformers, Diffusers, Groq, Gemini, REST APIs

•Kriti Support AI

Enterprise RAG Helpdesk powered by Local SLMs

GitHub

May 2026

- Architected a dual-dashboard **RAG pipeline** automating rulebook-bound query resolution for customer service.
- Implemented local SLM inference utilizing **TinyLlama via Ollama** with strict prompt-based boundary constraints to eliminate hallucinations.
- Built a localized indexing pipeline to ingest unstructured policy PDFs into a persistent **FAISS vector store**.
- **Technologies:** TinyLlama, Ollama, FAISS, HuggingFace (all-MiniLM-L6-v2), Flask Blueprint, Python, React.js

•PsycheTrack

Multimodal Health Analysis in Social Media

GitHub

Aug 2024 – Sep 2024

- Fine-tuned an on-device **Phi-3.5 SLM** via **QLoRA** in PyTorch to perform clinical risk stratification and output explainable reasoning for detected signs of depression.
- Engineered a multimodal feature-fusion pipeline integrating text NLP, audio MFCCs, and visual CNNs, boosting accuracy to **92% text, 90% audio, and 88% visual accuracy**.
- **Technologies:** Python, PyTorch, Hugging Face, QLoRA, Phi-3.5, TensorFlow, Keras, Librosa, OpenCV, Scikit-learn

•Athletiscan

Sports Celebrity Classification System (92% Accuracy)

GitHub

Feb 2024 – April 2024

- Implemented Haar cascade classifiers to isolate facial regions containing two or more eyes before classification.
- Synthesized raw image pixels with wavelet-transformed features to optimize feature extraction.
- Trained an SVM classifier on 1,000+ images; **reduced misclassification by 50%** using wavelet-enhanced features.
- **Technologies:** Python, OpenCV, Scikit-learn, pandas, NumPy, Joblib, Haar Cascades, Wavelet Transforms, SVM

Technical Skills and Interests

Languages: Python, Java, SQL, HTML, CSS

Libraries: PyTorch, TensorFlow, Hugging Face, LangChain, Diffusers, Librosa, OpenCV, Scikit-learn, NumPy

AI & Model Engineering: GenAI, LLMs/SLMs (Pre-training, Fine-tuning), QLoRA, RAG Pipelines, Transformer

Inference & Deployment: Flask, Streamlit, REST APIs, Git Control

Databases: MySQL, SQLite; **Vector DBs:** FAISS, Chroma

Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, LangSmith

Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering

Certifications

- Python for Data Science, AI & Development
- Data Science, ML, DL, NLP Bootcamp
- Gen AI Foundational Models for NLP
- Python Project for Data Science
- Generative AI: Introduction & Applications
- Gen AI: Language Modeling with Transformers

Achievements and Awards

- **LeetCode:** Solved 250+ questions in Java
- **HackerRank:** 5-star in Data Structures & Algorithms
- Solved 500+ coding problems across multiple platforms (LeetCode, HackerRank)